

Actividad GUÍA DE APRENDIZAJE – SEMANA 9

Introducción a Kubernetes

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Actividad de aprendizaje 1. Verificar el entorno y el acceso al cluster

```
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl version --client
Client Version: v1.36.2
Kustomize Version: v5.8.1
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl cluster-info
Kubernetes control plane is running at https://127.0.0.1:59046
CoreDNS is running at https://127.0.0.1:59046/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get nodes
NAME        STATUS    ROLES    AGE   VERSION
minikube    Ready     control-plane  3m52s  v1.35.1
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes#
```

Con los comandos ejecutados se puede visualizar la versión de kubernetes, la información de que kubernetes está corriendo correctamente, da información sobre el host y por último un estado de kubernetes.

Actividad de aprendizaje 2. Reconocer recursos básicos del cluster

```
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl version --client
Client Version: v1.36.2
Kustomize Version: v5.8.1
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl cluster-info
Kubernetes control plane is running at https://127.0.0.1:59046
CoreDNS is running at https://127.0.0.1:59046/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get nodes
NAME        STATUS    ROLES    AGE   VERSION
minikube    Ready     control-plane  3m52s  v1.35.1
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get namespaces
error: the server doesn't have a resource type "namespaces"
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get namespaces
NAME                STATUS    AGE
default             Active    10m
kube-node-lease     Active    10m
kube-public         Active    10m
kube-system         Active    10m
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get deployments
No resources found in default namespace.
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get deployments
No resources found in default namespace.
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get pods
No resources found in default namespace.
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get services
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE
kubernetes          ClusterIP   10.96.0.1     <none>         443/TCP    14m
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes#
```

Actividad de aprendizaje 3. Crear un namespace de laboratorio

```
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get namespaces
error: the server doesn't have a resource type "namespaces"
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get namespaces
NAME                STATUS    AGE
default             Active    10m
kube-node-lease     Active    10m
kube-public         Active    10m
kube-system         Active    10m
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get deployments
No resources found in default namespace.
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get deployments
No resources found in default namespace.
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get pods
No resources found in default namespace.
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get services
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE
kubernetes          ClusterIP   10.96.0.1     <none>         443/TCP    14m
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl create namespace laboratorio-k8s
namespace/laboratorio-k8s created
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get namespaces
NAME                STATUS    AGE
default             Active    19m
kube-node-lease     Active    19m
kube-public         Active    19m
kube-system         Active    19m
laboratorio-k8s     Active    19s
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl config set-context --current --namespace=laboratorio-k8s
Context "minikube" modified.
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes#
```

Actividad de aprendizaje 4. Crear el primer Deployment paso a paso

```

root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get deployments
No resources found in default namespace.
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get pods
No resources found in default namespace.
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get services
NAME      TYPE      CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl create namespace laboratorio-k8s
namespace/laboratorio-k8s created
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get namespaces
NAME      STATUS   AGE
default   Active   19m
kube-node-lease   Active   19m
kube-public   Active   19m
kube-system   Active   19m
laboratorio-k8s   Active   19s
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl config set-context --current --namespace=laboratorio-k8s
Context "minikube" modified.
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl create deployment web-nginx --image=nginx -n laboratorio-k8s
deployment.apps/web-nginx created
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get deployments -n laboratorio-k8s
No resources found in laboratorio-k8s namespace.
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get deployments -n laboratorio-k8s
NAME      READY   UP-TO-DATE   AVAILABLE   AGE
web-nginx  1/1     1             1           74s
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get pods -n laboratorio-k8s
NAME      READY   STATUS    RESTARTS   AGE
web-nginx-57bfd454d-qltqp  1/1     Running   0           114s
root@DESKTOP-4KJGQ9E: /mnt/c/users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes#

```

Actividad de aprendizaje 5. Inspeccionar el Deployment y los Pods

```

root@DESKTOP-4KJGQ9E: /mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes
root@DESKTOP-4KJGQ9E: /mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl describe deployment web-nginx -n laboratorio-k8s
Name:          web-nginx
Namespace:     laboratorio-k8s
CreationTimestamp:  Wed, 24 Jun 2026 21:39:09 -0500
Labels:        app=web-nginx
Annotations:    deployment.kubernetes.io/revision: 1
Selector:      app=web-nginx
Replicas:      1 desired | 1 updated | 1 total | 1 available | 0 unavailable
StrategyType:  RollingUpdate
MinReadySeconds:  0
RollingUpdateStrategy:  25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=web-nginx
  Containers:
    nginx:
      Image:          nginx
      Port:           <none>
      Host Port:      <none>
      Environment:    <none>
      Mounts:          <none>
      Volumes:         <none>
      Node-Selectors:  <none>
      Tolerations:     <none>
Conditions:
  Type           Status  Reason
  ----           -
  Progressing    True    NewReplicaSetAvailable
  Available      True    MinimumReplicasAvailable
OldReplicaSets: <none>
NewReplicaSet:  web-nginx-57bfd454d (1/1 replicas created)
Events:
  Type      Reason          Age    From          Message
  ----      -
  Normal    ScalingReplicaSet  9h    deployment-controller    Scaled up replica set web-nginx-57bfd454d from 0 to 1
root@DESKTOP-4KJGQ9E: /mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get pods -o wide -n laboratorio-k8s
NAME      READY   STATUS    RESTARTS   AGE   IP            NODE             NOMINATED NODE   READINESS GATES
web-nginx-57bfd454d-qltqp  1/1     Running   1 (9h ago)  9h    10.244.0.5    minikube        <none>            <none>
root@DESKTOP-4KJGQ9E: /mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl describe pod <nombre-del-pod> -n laboratorio-k8s
root@DESKTOP-4KJGQ9E: /mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl describe pod web-nginx-57bfd454d-qltqp -n laboratorio-k8s
Name:          web-nginx-57bfd454d-qltqp
Namespace:     laboratorio-k8s
Priority:       0
Service Account:  default
Node:          minikube/192.168.49.2

```

```

root@DESKTOP-4KJGQ9E: /mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes
Node:
  minikube/192.168.49.2
Start Time:
  Wed, 24 Jun 2026 21:39:09 -0500
Labels:
  app=web-nginx
  pod-template-hash=57bfdb454d
Annotations:
  Status:
    Running
IP:
  10.244.0.5
IPs:
  IP:
    10.244.0.5
Controlled By:
  ReplicaSet/web-nginx-57bfdb454d
Containers:
  nginx:
    Container ID:
      docker://a0f1462ffc64b04b67456b50f3334edc8a9ac620bd3ce755df366c31d859a995
    Image:
      nginx
    Image ID:
      docker-pullable://nginx@sha256:ec4ed8b5299e5e90694af7750eb6dff2627317d30544d056b0
    Port:
      <none>
    Host Port:
      <none>
    State:
      Running
    Started:
      Thu, 25 Jun 2026 07:21:33 -0500
    Last State:
      Terminated
    Reason:
      Completed
    Exit Code:
      0
    Started:
      Wed, 24 Jun 2026 21:39:28 -0500
    Finished:
      Wed, 24 Jun 2026 21:50:06 -0500
    Ready:
      True
    Restart Count:
      1
    Environment:
      <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-jk8cz (ro)
Conditions:
  Type
  PodReadyToStartContainers
  Initialized
  Ready
  ContainersReady
  PodScheduled
  Status
  True
  True
  True
  True
  True
Volumes:
  kube-api-access-jk8cz:
    Type:
      Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:
      3607
    ConfigMapName:
      kube-root-ca.crt
    Optional:
      false
    DownwardAPI:
      true
QoS Class:
  BestEffort

```

```

root@DESKTOP-4KJGQ9E: /mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes
QoS Class:
  BestEffort
Node-Selectors:
  <none>
Tolerations:
  node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
  node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type
  Reason
  Age
  From
  Message
  Normal
  Scheduled
  9h
  default-scheduler
  Successfully assigned laboratorio-k8s/web-nginx-57bfdb454d-qltvp to minikube
  Normal
  Pulling
  9h
  kubelet
  spec.containers(nginx): Pulling image "nginx"
  Normal
  Pulled
  9h
  kubelet
  spec.containers(nginx): Successfully pulled image "nginx" in 18.291s (18.291s including waiting). Image size: 1613081
  34 bytes.
  Normal
  Created
  9h
  kubelet
  spec.containers(nginx): Container created
  Normal
  Started
  9h
  kubelet
  spec.containers(nginx): Container started
  Normal
  SandboxChanged
  13m
  kubelet
  Pod sandbox changed, it will be killed and re-created.
  Normal
  Pulling
  13m
  kubelet
  spec.containers(nginx): Pulling image "nginx"
  Normal
  Pulled
  13m
  kubelet
  spec.containers(nginx): Successfully pulled image "nginx" in 3.589s (3.589s including waiting). Image size: 161308134
  bytes.
  Normal
  Created
  13m
  kubelet
  spec.containers(nginx): Container created
  Normal
  Started
  13m
  kubelet
  spec.containers(nginx): Container started
root@DESKTOP-4KJGQ9E: /mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes#

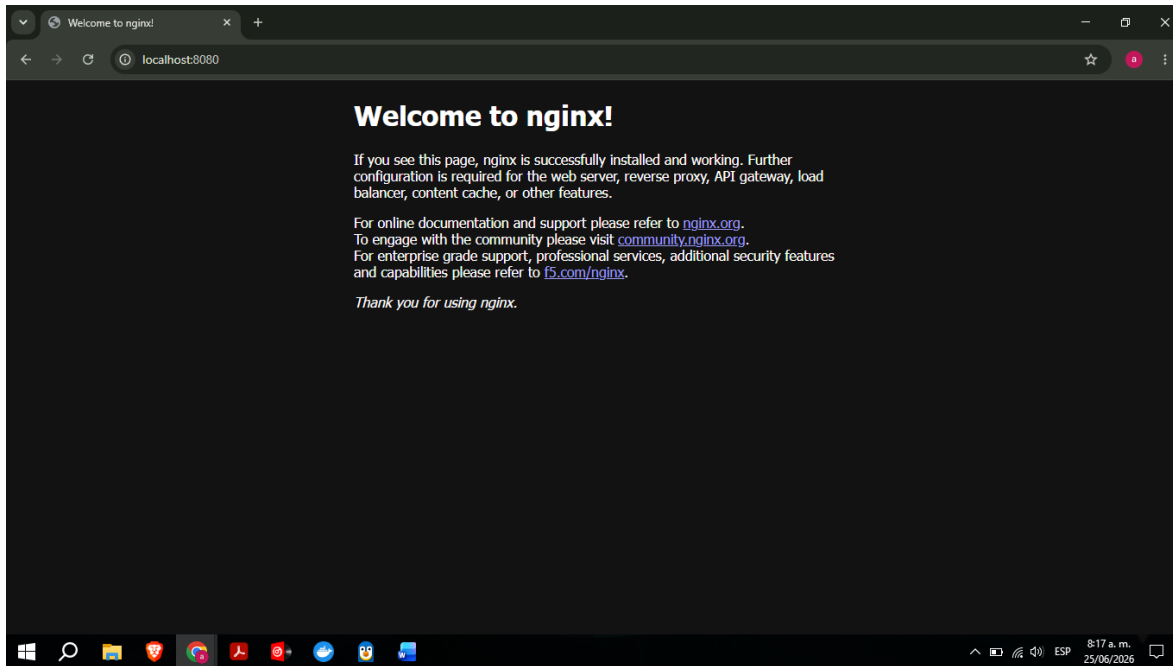
```

Actividad de aprendizaje 6. Exponer la aplicación con un Service y acceder localmente

```

root@DESKTOP-4KJGQ9E: /mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes
Normal
Pulled
9h
kubelet
spec.containers(nginx): Successfully pulled image "nginx" in 18.291s (18.291s including waiting). Image size: 1613081
34 bytes.
Normal
Created
9h
kubelet
spec.containers(nginx): Container created
Normal
Started
9h
kubelet
spec.containers(nginx): Container started
Normal
SandboxChanged
13m
kubelet
Pod sandbox changed, it will be killed and re-created.
Normal
Pulling
13m
kubelet
spec.containers(nginx): Pulling image "nginx"
Normal
Pulled
13m
kubelet
spec.containers(nginx): Successfully pulled image "nginx" in 3.589s (3.589s including waiting). Image size: 161308134
bytes.
Normal
Created
13m
kubelet
spec.containers(nginx): Container created
Normal
Started
13m
kubelet
spec.containers(nginx): Container started
root@DESKTOP-4KJGQ9E: /mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl expose deployment web-nginx --port=80 --type=ClusterIP -n laboratorio-k8s
service/web-nginx exposed
root@DESKTOP-4KJGQ9E: /mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get services -n laboratorio-k8s
NAME      TYPE      CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE
web-nginx ClusterIP    10.111.133.151 <none>         80/TCP      32s
root@DESKTOP-4KJGQ9E: /mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl port-forward service/web-nginx 8080:80 -n laboratorio-k8s
error: unable to listen on any of the requested ports: [[8080 80]]
root@DESKTOP-4KJGQ9E: /mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# ^C
root@DESKTOP-4KJGQ9E: /mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# docker stop ecoverde_web ecoverde_db
ecoverde_web
ecoverde_db
root@DESKTOP-4KJGQ9E: /mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl port-forward service/web-nginx 8080:80 -n laboratorio-k8s
Forwarding from 127.0.0.1:8080 -> 80
Forwarding from [::]:8080 -> 80
Handling connection for 8080
Handling connection for 8080

```



Actividad de aprendizaje 7. Escalar el Deployment

```
root@DESKTOP-4KJGQ9E:/mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes
root@DESKTOP-4KJGQ9E:/mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl expose deployment web-nginx --port=80 --type=ClusterIP -n laboratorio-k8s
service/web-nginx exposed
root@DESKTOP-4KJGQ9E:/mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get services -n laboratorio-k8s
NAME         TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
web-nginx    ClusterIP   10.111.133.151 <none>        80/TCP      32s
root@DESKTOP-4KJGQ9E:/mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl port-forward service/web-nginx 8080:80 -n laboratorio-k8s
Unable to listen on port 8080: Listeners failed to create with the following errors: [unable to create listener: Error listen tcp4 127.0.0.1:8080: bind: address already
in use unable to create listener: Error listen tcp6 [::]:8080: bind: address already in use]
error: unable to listen on any of the requested ports: [[8080 80]]
root@DESKTOP-4KJGQ9E:/mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# ^C
root@DESKTOP-4KJGQ9E:/mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# ^C
root@DESKTOP-4KJGQ9E:/mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# docker stop ecoverde_web ecoverde_db
ecoverde_web
ecoverde_db
root@DESKTOP-4KJGQ9E:/mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl port-forward service/web-nginx 8080:80 -n laboratorio-k8s
Forwarding from 127.0.0.1:8080 -> 80
Forwarding from [::]:8080 -> 80
Handling connection for 8080
Handling connection for 8080
^Croot@DESKTOP-4KJGQ9E:/mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl scale deployment/web-nginx --replicas=3 -n laboratorio-k8s
deployment.apps/web-nginx scaled
root@DESKTOP-4KJGQ9E:/mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get deployments -n laboratorio-k8s
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
web-nginx     3/3     3             3           10h
root@DESKTOP-4KJGQ9E:/mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get pods -n laboratorio-k8s
NAME                                READY   STATUS    RESTARTS   AGE
web-nginx-57bfdb454d-dbhqr          1/1     Running   0           82s
web-nginx-57bfdb454d-pxjs4          1/1     Running   0           82s
web-nginx-57bfdb454d-qlotp          1/1     Running   1 (10h ago) 11h
root@DESKTOP-4KJGQ9E:/mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes#
```

Actividad de aprendizaje 8. Aplicar recursos desde YAML

```
root@DESKTOP-4KJGQ9E: /mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes
> replicas: 2
> selector:
>   matchLabels:
>     app: demo-yaml
> template:
>   metadata:
>     labels:
>       app: demo-yaml
>   spec:
>     containers:
>     - name: nginx
>       image: nginx
>     ports:
>     - containerPort: 80
F> EOF
root@DESKTOP-4KJGQ9E: /mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl apply -f deployment-demo.yaml
deployment.apps/demo-yaml created
root@DESKTOP-4KJGQ9E: /mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get deployments -n laboratorio-k8s
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
demo-yaml     2/2     2            2           39s
web-nginx    3/3     3            3           11h
root@DESKTOP-4KJGQ9E: /mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get pods -n laboratorio-k8s
NAME          READY   STATUS    RESTARTS   AGE
demo-yaml-54cc79d7f-kgh7s  1/1     Running   0           61s
demo-yaml-54cc79d7f-vfdfp  1/1     Running   0           61s
web-nginx-57bfdb454d-dbhqr  1/1     Running   0           34m
web-nginx-57bfdb454d-pxjs4  1/1     Running   0           34m
web-nginx-57bfdb454d-qlotp  1/1     Running   1 (11h ago)  11h
root@DESKTOP-4KJGQ9E: /mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes#
```

Actividad de aprendizaje 9. Eliminar recursos de práctica

```
root@DESKTOP-4KJGQ9E: /mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes
>   image: nginx
>   ports:
>   - containerPort: 80
F> EOF
root@DESKTOP-4KJGQ9E: /mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl apply -f deployment-demo.yaml
deployment.apps/demo-yaml created
root@DESKTOP-4KJGQ9E: /mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get deployments -n laboratorio-k8s
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
demo-yaml     2/2     2            2           39s
web-nginx    3/3     3            3           11h
root@DESKTOP-4KJGQ9E: /mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl get pods -n laboratorio-k8s
NAME          READY   STATUS    RESTARTS   AGE
demo-yaml-54cc79d7f-kgh7s  1/1     Running   0           61s
demo-yaml-54cc79d7f-vfdfp  1/1     Running   0           61s
web-nginx-57bfdb454d-dbhqr  1/1     Running   0           34m
web-nginx-57bfdb454d-pxjs4  1/1     Running   0           34m
web-nginx-57bfdb454d-qlotp  1/1     Running   1 (11h ago)  11h
root@DESKTOP-4KJGQ9E: /mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl delete deployment web-nginx -n laboratorio-k8s
deployment.apps "web-nginx" deleted from laboratorio-k8s namespace
root@DESKTOP-4KJGQ9E: /mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl delete service web-nginx -n laboratorio-k8s
service "web-nginx" deleted from laboratorio-k8s namespace
root@DESKTOP-4KJGQ9E: /mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl delete deployment demo-yaml -n laboratorio-k8s
deployment.apps "demo-yaml" deleted from laboratorio-k8s namespace
root@DESKTOP-4KJGQ9E: /mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes# kubectl delete namespace laboratorio-k8s
namespace "laboratorio-k8s" deleted
root@DESKTOP-4KJGQ9E: /mnt/c/Users/dmk/desktop/SENA/DEVOPS Y DOCKERS/Kubernetes#
```

Actividad de aprendizaje 10. Cierre y reflexión

Responder de forma breve:

- ¿Qué diferencia encontré entre contenedor Docker y Pod de Kubernetes?
El contenedor de Docker es la aplicación empaquetada con todo lo necesario para correr sola, mientras que el Pod es la unidad de Kubernetes que actúa como una "envoltura" o casita para ese contenedor, permitiendo que el clúster lo controle y administre de forma nativa.
- ¿Para qué sirve un Deployment en un cluster?
Sirve como un supervisor automático de la aplicación. Su trabajo principal es garantizar que el número de copias de la aplicación que definimos esté siempre activo; si un contenedor falla o se cae, el Deployment lo detecta y lo vuelve a levantar de inmediato sin que el usuario lo note.
- ¿Qué ventaja aporta un Service frente a conectarse directo a un Pod?

La gran ventaja es la estabilidad de la conexión. Como los Pods son efímeros y cambian de dirección IP cada vez que se reinician, el Service funciona como una puerta de acceso fija y única que distribuye el tráfico, evitando que la conexión se rompa si un Pod se destruye.

- ¿Qué importancia tiene el namespace en un ambiente compartido?

Es fundamental para el orden y la seguridad. Funciona como una división o "barrera virtual" dentro del clúster que permite separar proyectos o equipos de trabajo (por ejemplo, entornos de desarrollo y producción), evitando que los recursos de una actividad se mezclen o interfieran con otra.

- ¿Qué comando me resultó más útil durante la práctica?}

A nivel de diagnóstico, kubectl describe fue el más útil porque permite "abrir el capó" de cualquier recurso para entender detalladamente qué está pasando por dentro, revisar su estado y corregir errores cuando algo no arranca como se espera.